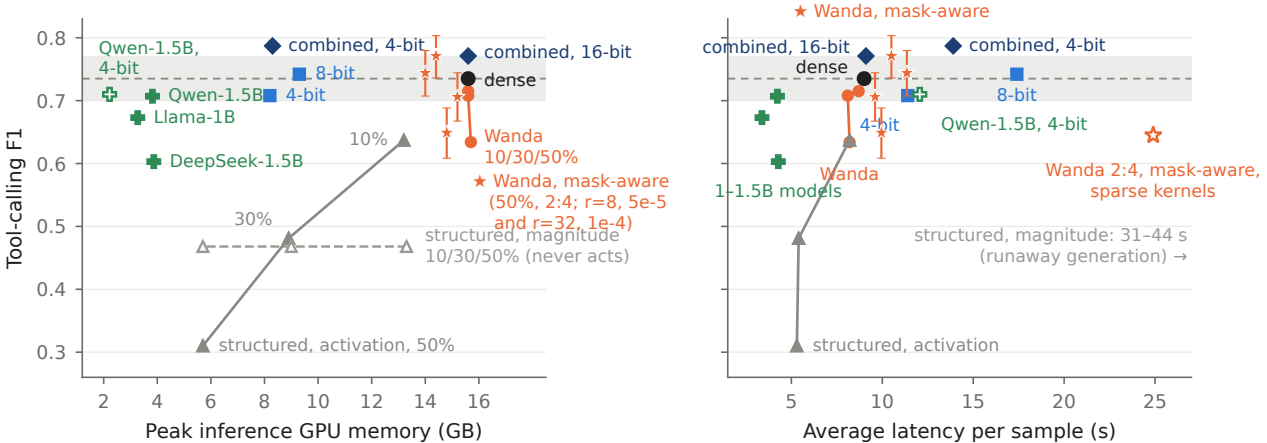
Tool-calling F1
0.8
0.7
0.6
0.5
0.4
0.3
Qwen-1.5B, 4-bit
combined, 4-bit
combined, 16-bit
8-bit
dense
Qwen-1.5B
4-bit
Llama-1B
DeepSeek-1.5B
10%
Wanda 10/30/50%
★ Wanda, mask-aware (50%, 2:4; r=8, 5e-5 and r=32, 1e-4)
30%
structured, magnitude 10/30/50% (never acts)
structured, activation, 50%
2
4
6
8
10
12
14
16
Peak inference GPU memory (GB)
★ Wanda, mask-aware
combined, 16-bit
combined, 4-bit
dense
8-bit
4-bit
Qwen-1.5B, 4-bit
Wanda 2:4, mask-aware, sparse kernels
Wanda
1-1.5B models
structured, magnitude: 31–44 s (runaway generation) →
structured, activation
5
10
15
20
25
Average latency per sample (s)